

12.7 cm

LF-0216

DUODIN
(Injection)

Description: It is clear, colorless, sterile solution for injection.

Composition: Each 2 mL contains Cimetidine 200 mg.

Pharmacology/Pharmacokinetics:

Pharmacodynamics:

Cimetidine competitively inhibits the action of histamine at the histamine H₂ receptors of the parietal cells and thus is histamine H₂-receptor antagonist. Cimetidine is not an anticholinergic agent. Studies have shown that Cimetidine inhibits both daytime and nocturnal basal gastric acid secretion. Cimetidine also inhibits gastric acid secretion stimulated by food, histamine, pentagastrin, caffeine and insulin. Cimetidine indirectly reduces pepsin secretion by decreasing the volume of gastric juice. The drug augments the normal serum gastrin increase in response to a meal but does not increase nocturnal serum gastrin concentrations in fasting patients. Cimetidine appears to inhibit betazole and pentagastrin-stimulated rise in intrinsic factor concentration. Cimetidine has no substantial effect on lower esophageal sphincter pressure, gastric motility or emptying, or biliary or pancreatic secretion.

Pharmacokinetics:

Cimetidine parenteral (IV or IM) administration provide a period of therapeutically effective blood levels; blood concentrations remain above that required to provide 80% inhibition of basal gastric secretion for 4 to 5 hours following a dose of 300 mg. Steady-state plasma cimetidine concentrations achieved following IV infusion of the drug depend on the clearance of the drug but are proportional to the rate of infusion. In one study in patients with peptic ulcer and normal renal function, IV infusion of the drug at 37.5 mg/hour resulted in steady-state plasma cimetidine concentrations of 0.9 mg/mL.

Cimetidine is widely distributed throughout the body and is 15-20% bound to plasma proteins. Animal studies indicate that the drug crosses the placenta. Cimetidine is distributed into milk. Cimetidine is metabolized in the liver to sulfoxide and 5-hydroxymethyl derivatives, and possibly guanylurea, although this latter compound may result from in vitro degradation. The drug has a blood half-life of about 2 hours in patients with normal renal function, 2.9 hours in patients with creatinine clearances of 20-50 mL/minute, 3.7 hours in patients with creatinine clearances of less than 20 mL/minute, and 5 hours in anephric patients. Following IV administration of the drug in critically ill children 4.1-15 years of age, plasma cimetidine concentrations appeared to decline in a biphasic manner, with a mean distribution half-life (t_{1/2α}) of about 10 minutes and a mean elimination half-life (t_{1/2β}) of about 1.4 hours; plasma cimetidine sulfoxide concentration also appeared to decline in a biphasic manner with t_{1/2α} of 22 minutes and a t_{1/2β} of 2.6 hours.

The principal route of excretion of Cimetidine is the urine. Following parenteral administration, most of the drug is excreted as the parent compound; Following IV or IM administration, approximately 75% of the drug is recovered from the urine after 24 hours as the parent compound.

Studies show that following IV administration of radiolabeled cimetidine, 80-90% of the drug is excreted in urine within 24 hours; 50-73% is excreted unchanged and the remainder as the two metabolites. About 10% of the drug is excreted in feces. In one study, cimetidine was completely removed from the circulation after 5 hours of hemodialysis. (Ref : AHFS Drug Information 92, p. 1755)

Indications: It is indicated for the treatment of duodenal and benign gastric ulceration, including that associated with non-steroidal anti-inflammatory agents, recurrent and stomal ulceration, esophageal reflux disease and other condition where reduction of Cimetidine has shown benefit, reduction of gastric acid and pepsin secretion.

Contraindications: Cimetidine is contraindicated for patients known to have hypersensitivity to the product.

Side Effects/Adverse Reactions: Adverse effects reported in patients taking Cimetidine are described below by body system. Incidence

figures of 1 in 100 and greater are generally derived from controlled clinical studies.

Gastrointestinal: Diarrhea (usually mild) has been reported in approximately 1 in 100 patients.

CNS: Headaches, ranging from mild to severe, have been reported in 3.5% of 924 patients taking 1600 mg/day, 2.1% of 2,225 patients taking 800 mg/day and 2.3% of 1,897 patients taking placebo. Dizziness and somnolence (usually mild) have been reported in approximately 1 in 100 patients on either 1600 mg/day or 800 mg/day.

Reversible confusional states, e.g., mental confusion, agitation, psychosis, depression, anxiety, hallucinations, disorientation, usually in the elderly or already very ill patients, e.g. those with renal failure, have been reported predominantly, but not exclusively. They have usually developed within 2 to 3 days of initiation of Cimetidine therapy and have cleared within 3 to 4 days of discontinuation of the drug.

Endocrine: Gynecomastia has been reported in patients treated for 1 month or longer. In patients being treated for pathological hypersecretory states, this occurred in about 4% of cases while in all others the incidence was 0.3% to 1% in various studies. No evidence of induced endocrine dysfunction was found, and the condition remained unchanged or returned toward normal with continuing Cimetidine treatment.

Reversible impotence has been reported in patients with pathological hypersecretory disorders, e.g. Zollinger-Ellison Syndrome, receiving Cimetidine, particularly in high doses, for at least 12 months (range 12 to 79 months, mean 38 months). However, in large-scale surveillance studies at regular dosage, the incidence has not exceeded that commonly reported in the general population.

Hematology: Leucopenia, decreased white blood cell counts in Cimetidine-treated patients (approximately 1 per 100,000 patients), including agranulocytosis (approximately 3 per million patients), have been reported, including a few reports of recurrence on rechallenge. Most of these reports were in patients who had serious concomitant illnesses and received drugs and/or treatment known to produce neutropenia.

Thrombocytopenia (approximately 3 per million patients) and, very rarely, cases of pancytopenia or aplastic anemia have also been reported. As with some other H₂-receptor antagonists, there have been extremely rare reports of immune hemolytic anemia.

Hepatobiliary: Dose-related increases in serum transaminase have been reported. In most cases they did not progress with continued therapy and returned to normal at the end of therapy. There have been rare reports of cholestatic or mixed cholestatic-hepatocellular effects. These were usually reversible. Because of the predominance of cholestatic features, severe parenchymal injury is considered highly unlikely. However, as in the occasional liver injury with other H₂-receptor antagonists, in exceedingly rare circumstances fatal outcomes have been reported.

There has been reported a single case of biopsy-proven periportal hepatic fibrosis in a patient receiving Cimetidine.

Rare cases of pancreatitis, which cleared on withdrawal of the drug, have been reported.

Hypersensitivity: Rare cases of fever and allergic reactions including anaphylaxis and hypersensitivity vasculitis, which cleared on withdrawal of the drug, have been reported.

Renal: Small, possibly dose-related increases in plasma creatinine, presumably due to competition for renal tubular secretion, are not uncommon and do not signify deteriorating renal function. Rare cases of interstitial nephritis and urinary retention, which cleared on withdrawal of the drug, have been reported.

Cardiovascular: Rare cases of bradycardia, tachycardia and A-V heart block have been reported with H₂-receptor antagonists.

Musculoskeletal: There have been rare reports of reversible arthralgia and myalgia: exacerbation of joint symptoms in patients with preexisting arthritis has also been reported. Such symptoms have usually been alleviated by a reduction in Cimetidine dosage. Rare cases of polymyositis have been reported, but no causal relationship has been established.

Integumental: Mild rash and, very rarely, cases of severe generalized skin reactions including Stevens-Johnson syndrome, epidermal necrolysis, erythema multiforme, exfoliative dermatitis and generalized exfoliative erythroderma have been reported with H₂-receptor antagonists. Reversible alopecia has been reported very rarely. (Ref : PDR Generics 1995, p. 618)

Precautions/Warnings: General: Rare instances of cardiac arrhythmias and hypotension have been reported following the rapid administration of Cimetidine hydrochloride Injection by intravenous bolus.

Symptomatic response to Cimetidine therapy does not preclude the presence of a gastric malignancy. Cimetidine can mask the symptoms and allow transient healing of gastric cancer. There have been rare reports of transient healing of gastric ulcers despite subsequently documented malignancy. Reversible confusional states (see "Adverse Reactions") have been observed on occasion, predominantly, but not exclusively, in severely ill patients. Advancing age (50 or more years) and preexisting liver and/or renal disease appear to be contributing factors. In some patients these confusional states have been mild and have not required discontinuation of Cimetidine therapy. In cases where discontinuation was judged necessary, the condition usually cleared within 3 to 4 days of drug withdrawal.

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Dosage should be reduced in patients with impaired renal function according to creatinine clearance. The following dosages are suggested: creatinine clearance of 0 to 15 mL per minute, 200 mg twice a day; 15 to 30 mL per minute, 200 mg three times a day; 30 to 50 mL per minute, 200 mg four times a day; over 50 mL per minute, normal dosage. Cimetidine is removed by hemodialysis, but not to any extent by peritoneal dialysis.

Care should be taken to observe periodically patients on prolonged treatment.

Care should be taken that patients with a history of peptic ulcer, particularly an elderly being treated with Cimetidine and non-steroidal anti-inflammatory agent are observed regularly.

In patients on drug treatment or with illnesses that cause a fall in blood cell count, the possibility that H₂-receptor antagonism could potentiate this effect should be borne in mind.

Close monitoring of patients on Cimetidine receiving oral anti-coagulants or phenytoin is recommended and a reduction in the dosage of these drugs may be necessary.

Drug Interactions: Cimetidine, apparently through an effect on certain microsomal enzyme systems, has been reported to reduce the hepatic metabolism of warfarin-type anticoagulants, phenytoin, propranolol, nifedipine, chlorthalidoxepoxide, diazepam, certain tricyclic antidepressants, lidocaine, theophylline and metronidazole, thereby delaying elimination and increasing blood levels of these drugs.

Clinically significant effects have been reported with the warfarin anticoagulants; therefore, close monitoring of prothrombin time is recommended, and adjustment of the anticoagulant dose may be necessary when Cimetidine is administered concomitantly. Interaction with phenytoin, lidocaine and theophylline has also been reported to produce adverse clinical effects.

However, a crossover study in healthy subjects receiving either Cimetidine 300 mg q.i.d. or 800 mg h.s. concomitantly with a 300 mg b.i.d. dosage of theophylline demonstrated less alteration in steady-state theophylline peak serum levels with the 800 mg h.s. regimen, particularly in subjects aged 54 years and older. Data beyond 10 days are not available. (Note: All patients receiving theophylline should be monitored appropriately, regardless of concomitant drug therapy.)

Dosage of the drugs mentioned above and other similarly metabolized drugs, particularly those of low therapeutic ratio or in patients with renal and/or hepatic impairment, may require adjustment when starting or stopping concomitantly administered Cimetidine to maintain optimum therapeutic blood levels.

Additional clinical experience may reveal other drugs affected by the concomitant administration of Cimetidine.

Carcinogenesis, Mutagenesis, Impairment of Fertility: In a 24-month toxicity study conducted in rats, at dose levels of 150, 378 and 950 mg/kg/day (approximately 8 to 48 times the recommended human dose), there was a small increase in the incidence of benign Leydig cell tumors in each dose group; when the combined drug-treated groups and control groups were compared, this increase reached statistical significance. In a subsequent 24-month study, there were no differences between the rats receiving 150 mg/kg/day and the untreated controls. However, a statistically significant increase in benign Leydig cell tumor incidence was seen in the rats that received 378 and 950 mg/kg/day. These tumors were common in control groups as well as treated groups and the difference became apparent only in aged rats.

Cimetidine has demonstrated a weak antiandrogenic effect. In animal studies this was manifested as reduced prostate and seminal vesicle weights. However, there was no impairment of mating performance or fertility, nor any harm to the fetus in these animals at doses 8 to 48 times the full therapeutic dose of Cimetidine, as compared with controls. The cases of gynecomastia seen in patients treated for 1 month or longer may be related to this effect. In human studies Cimetidine has been shown to have no effect on spermatogenesis, sperm count, motility, morphology or in vitro fertilizing capacity.

Pregnancy: Teratogenic Effects. Pregnancy Category B: Reproduction studies have been performed in rats, rabbits and mice at doses up to 40 times the normal human dose and have revealed no evidence of impaired fertility or harm to the fetus due to Cimetidine. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproductive studies are not always predictive of human response, this drug should be used during pregnancy only if clearly needed.

Nursing Mothers: Cimetidine is secreted in human milk and, as a general rule, nursing should not be undertaken while a patient is on a drug.

Pediatric Precautions: Clinical experience in children is limited. Therefore, Cimetidine therapy cannot be recommended for children under 16, unless, in the judgment of the physician, anticipated benefits outweigh the potential risks. In very limited experience, doses of 20 to 40 mg/kg body weight per day have been used.

Dosage and Administration: Administer 200 mg by intramuscular every 4 hours.

The usual dosage for intravenous administration is 200-400 mg, which may be repeated four-to six-hourly. If IV injection cannot be avoided, 200 mg should be given slowly over at least 2 minutes, and may be repeated four-to-six hourly. Rapid IV injection has been associated with cardiac arrhythmias. If there is cardiovascular impairment, or if a larger dose is needed, the dose should be diluted and given over at least 10 minutes. In such cases infusion is preferable.

If continuous IV infusion is required, Cimetidine may be given at an average rate of 50 to 100 mg / hour over 24 hours.

The total daily dose by any route should not normally exceed 2.4 g. Dosage should be reduced in patients with impaired renal failure.

Pediatric Dosage: When the potential benefits are thought to outweigh the possible risks (See "Pediatric Precautions"), a pediatric cimetidine dosage of 20 mg – 40 mg per kg daily in divided doses has been used in a limited number of children.

Stability: Cimetidine has been shown to be compatible with electrolyte and dextrose solutions commonly used for intravenous infusion.

Incompatibilities: Cimetidine is compatible with the commonly used IV solutions such as sodium chloride 0.9%, glucose 5 or 10% and lactated Ringer's solution. The compatibility depends on several factors such as concentration, diluents used, pH, temperature. It is suggested to consult specialized references for more specific information.

Symptoms and Treatment for Overdosage:

Symptoms: Studies in animals indicate that toxic doses are associated with respiratory failure and tachycardia that may be controlled by assisted respiration and the administration of a beta-blocker. There have been reports of severe CNS symptoms, including unresponsiveness, following ingestion of between 20 and 40 grams of Cimetidine, and extremely rare reports following concomitant use of multiple CNS-active medications and ingestion of Cimetidine at doses less than 20 grams. An elderly, terminally ill dehydrated patient with organic brain syndrome receiving concomitant antipsychotic agents and Cimetidine 4800 mg intravenously over a 24-hour period experienced mental deterioration with reversal on Cimetidine discontinuation.

Treatment: For the treatment of cimetidine overdosage, the usual measures to remove unabsorbed drug from the GI tract, clinical monitoring, and supportive therapy should be employed.

Packing / Pack Size: 10 boxes x 10 ampoules per box x 2 mL per ampoule

Storage: Store below 30°C. Protect from light.

Shelf-life: 3 years

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